14-W1　回复力模型中的分子极化率

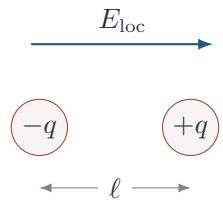
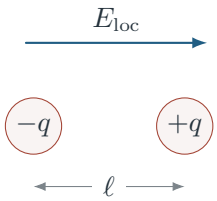


图 14-9　题图　外场使两电荷中心分离，回复力限制位移

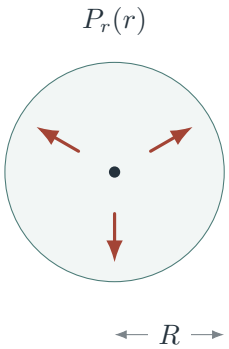


$$k\ell = qE$$

$$p = q\ell = q^2E/k$$

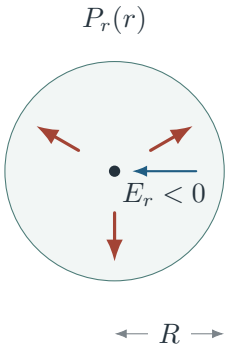
图 14-10　解答分析图　力平衡确定相对位移，偶极矩须再乘一次电荷

14-W2　径向极化球的体电荷与表面电荷



$$\boldsymbol{P} = \lambda \boldsymbol{r}$$

图 14-11　题图　径向极化在体内和球面留下异号电荷



$$\boldsymbol{P} = \lambda \boldsymbol{r}$$

$$\rho_b = -3\lambda$$

$$\sigma_b = \lambda R$$

$$E_{\text{out}} = 0$$

图 14-12　解答分析图　体电荷和表面电荷总量抵消，内场朝向球心

14-W3　均匀极化圆棒的轴线场

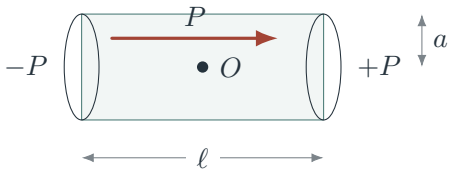
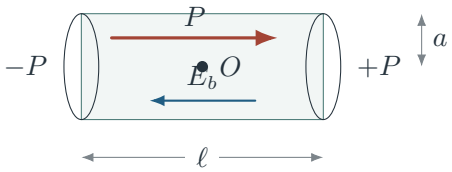


图 14-13　题图　端面带相反电荷，侧面法向投影为零



$s_{\pm} = \ell/2 \mp z$

中心弱，端部强

图 14-14　解答分析图　两圆盘在棒内部的轴线场均朝负轴方向

14-W4　均匀极化球的电势和自身能量

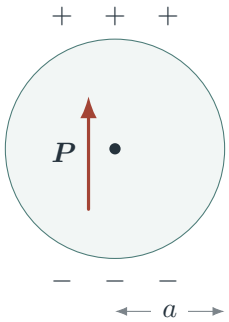
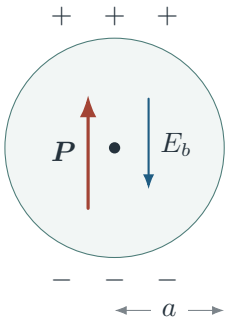


图 14-15　题图　球面极化电荷按极角余弦分布



$$p = 4\pi a^3 P/3$$
$$U_b = VP^2/(6\varepsilon_0)$$

图 14-16　解答分析图　球内极化场反向，球外还储存偶极场能

14-W5　均匀外场中的线性介质球

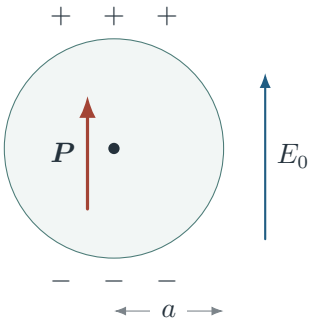
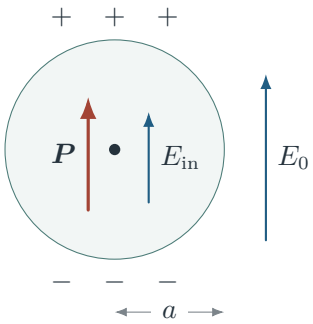


图 14-17　题图　介质球的外场方向与极化方向相同



$$E_{\text{in}} = 3E_0/(\kappa+2)$$
$$P = \varepsilon_0(\kappa-1)E_{\text{in}}$$

图 14-18　解答分析图　内场仍顺外场，但被球面极化电荷削弱

14-W6　电容器填入介质时的两种连接

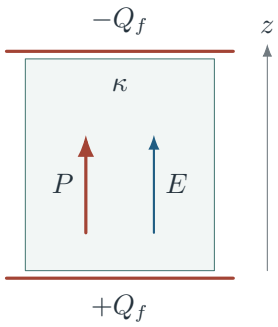
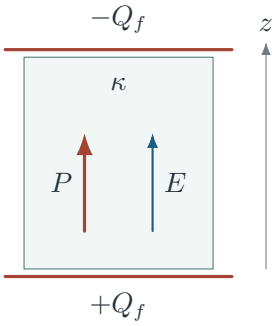


图 14-19　题图　介质填满极板间隙，箭头标出总场和极化



断开：Q_f 不变

接源：U 不变

$C = \kappa C_0$

图 14-20　解答分析图　连接决定固定电荷或固定电压，电容相同

14-W7　横向外场中的无限长介质圆柱

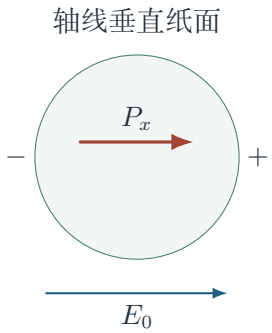
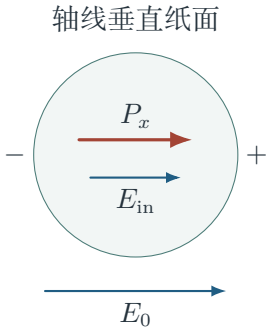


图 14-21　题图　横截面中，极化从负电荷侧指向正电荷侧



$$E_{\text{in}} = 2E_0/(\kappa+1)$$

$$\sigma_b = P\cos\theta$$

图 14-22　解答分析图　横向内场与极化同向，退极化因子为二分之一

14-W8　两层介质之间的极化面电荷

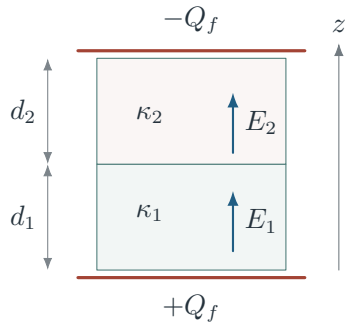
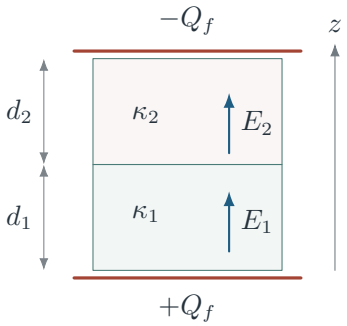


图 14-23　题图　两介质层沿场方向叠放，法向从下层指向上层



$$D_{1z} = D_{2z}$$

$$\sigma_{b,12} = P_1 - P_2$$

图 14-24　解答分析图　电位移法向连续，电场与极化可在界面跳变

14-W9　椭球的形状极化率与转矩

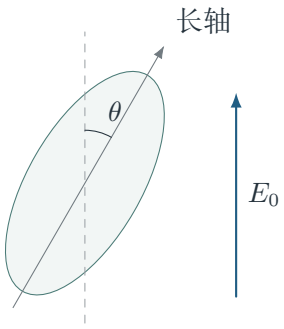


图 14-25　题图　材料各向同性，椭球长短轴仍有不同响应

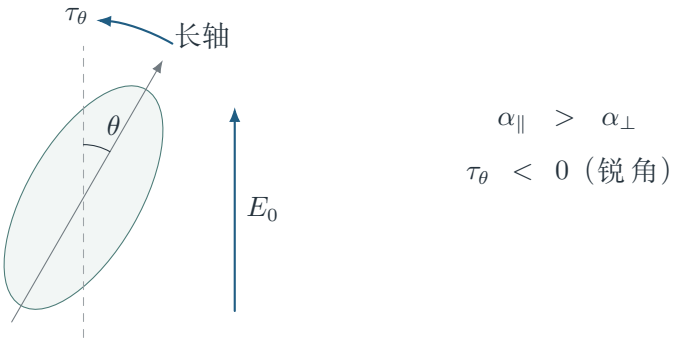


图 14-26　解答分析图　转矩减小长轴与外场的锐角，使长轴趋于顺场

14-W10　永久偶极子的取向饱和

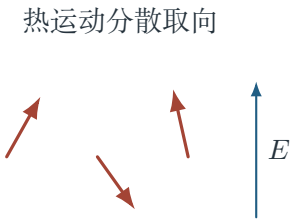


图 14-27　题图　永久偶极子在场和热运动共同作用下分布

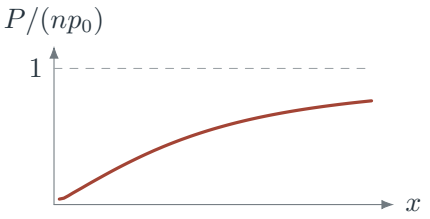


图 14-28　解答分析图　平均投影趋于分子偶极矩大小，不能无限增长

14-W11　真实球腔与冻结极化空腔

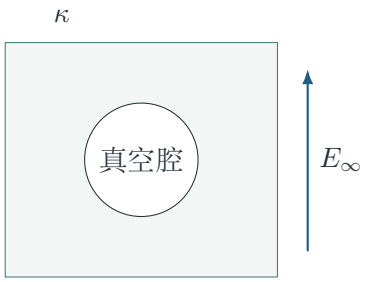


图 14-29　题图　真空球腔的界面没有自由面电荷

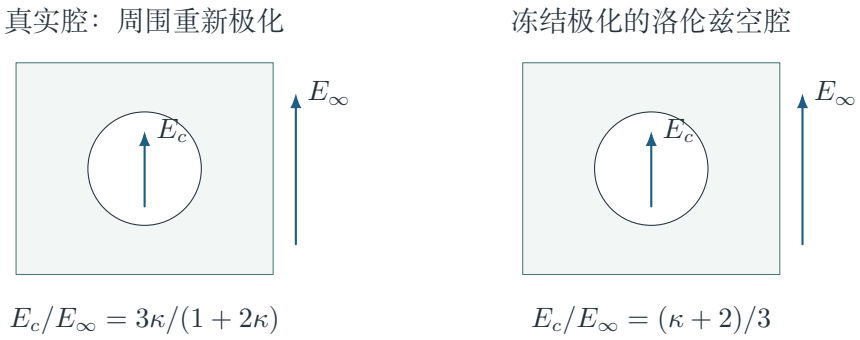
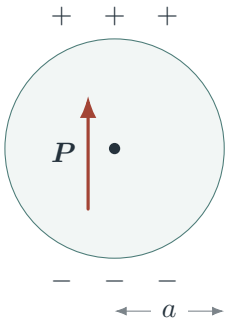


图 14-30　解答分析图　两种空腔采用不同的周围极化约束，不能混用腔内场系数

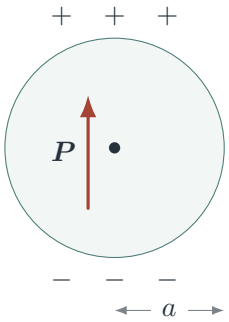
14-W12　退极化能与自洽极化



$N = 1/3$（球）

N_i 随形状改变

图 14-31　题图　给定几何因子后，极化电荷的自身场能可计算



$N = 1/3$（球）

N_i 随形状改变

内部能 + 退极化能

$G_{\min} = -\alpha E_0^2/2$

图 14-32　解答分析图　外场相互作用也须加入，极小化恢复自洽关系

14-T1　斜向极化的长方体

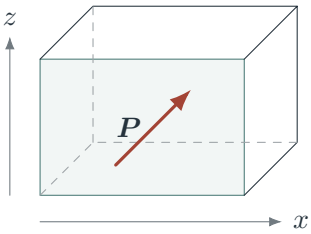
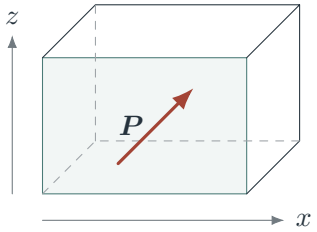


图 14-33　题图　均匀极化可斜向，六面法向须分别确定



$$\begin{aligned}\sigma_{x\pm} &= \pm P_x\\ \sigma_{y\pm} &= \pm P_y\\ \sigma_{z\pm} &= \pm P_z\end{aligned}$$

图 14-51　解答分析图　相对两面电荷抵消，总偶极矩仍可非零

14-T2　没有极化电荷的非均匀分布

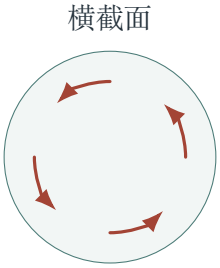
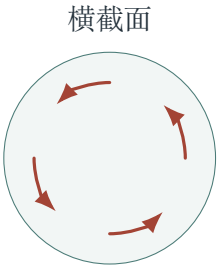


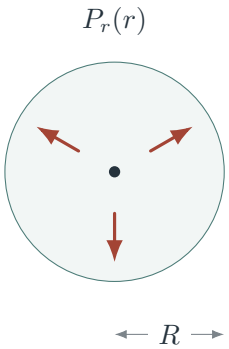
图 14-34　题图　极化沿方位绕行，处处切于侧面



$$\nabla \cdot \boldsymbol{P} = 0$$
$$\boldsymbol{P} \cdot \boldsymbol{n} = 0$$
$$\boldsymbol{E} = 0$$

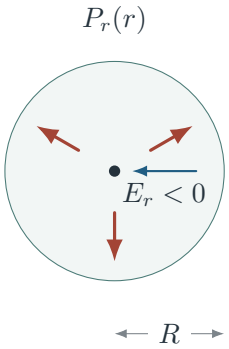
图 14-52　解答分析图　散度与所有边界法向分量均为零

14-T3　幂律径向极化的球



$$P_r = Cr^s$$

图 14-35　题图　极化幅值随半径作幂律变化



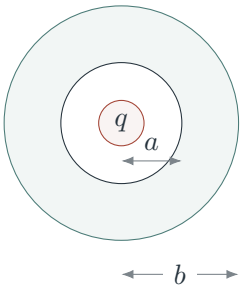
$$P_r = Cr^s$$

$$\rho_b = -C(s+2)r^{s-1}$$

$$\sigma_b = CR^s$$

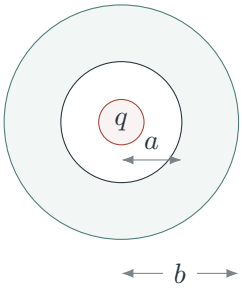
图 14-53　解答分析图　球面正电荷补偿体内负电荷，外场为零

14-T4　介电常数随半径变化的球壳



$$\kappa(r) = \kappa_0(1 + r/a)$$

图 14-36　题图　自由电荷在空腔中心，球壳的材料参数沿半径变化



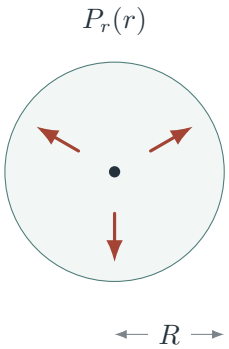
$$\kappa(r) = \kappa_0(1 + r/a)$$

$$4\pi r^2 D_r = q$$

$$\rho_b \propto -\kappa'/\kappa^2$$

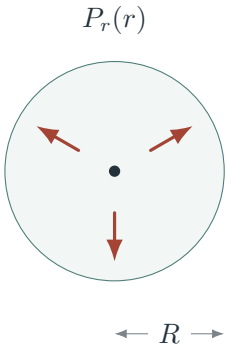
图 14-54　解答分析图　内表面法向朝球心，外表面法向朝外

14-T5　均匀材料中的非均匀极化



$$\rho_f = \text{const}$$
$$\kappa = \text{const}$$

图 14-37　题图　自由电荷均匀分布，极化却随半径增长



$$\rho_f = \text{const}$$
$$\kappa = \text{const}$$
$$\rho_b = -(1-1/\kappa)\rho_f$$
$$\sigma_b = (1-1/\kappa)\rho_f R/3$$

图 14-55　解答分析图　均匀材料中的非均匀场可产生极化体电荷

14-T6　斜向外场中的长圆柱

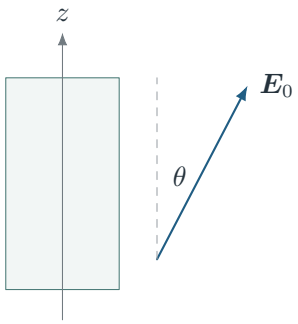
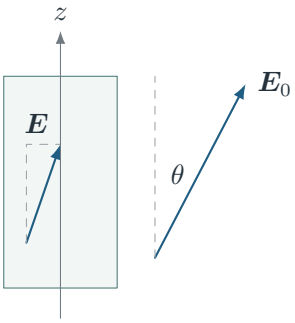


图 14-38　题图　图为轴向侧视，外场具有轴向和横向分量



$$E_{\parallel} = E_{0\parallel}$$
$$E_{\perp} = 2E_{0\perp}/(\kappa+1)$$

图 14-56　解答分析图　两个场分量受不同几何退极化作用

14-T7　部分插入介质的电容器

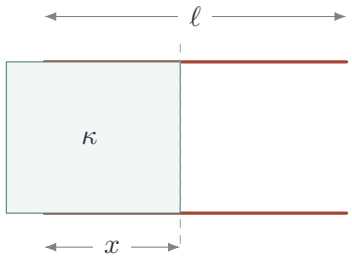
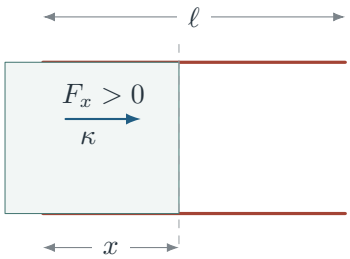


图 14-39　题图　介质向右插入，厚度填满极板间距



$$C' = \varepsilon_0 w(\kappa - 1)/d$$

$$F = U(x)^2 C'/2$$

图 14-57　解答分析图　电容随插入增加，机械力向内

14-T8　带永久极化片的电容器

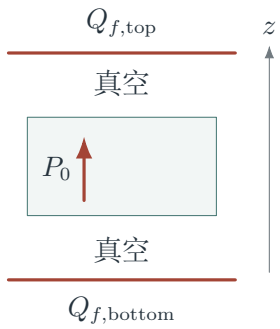
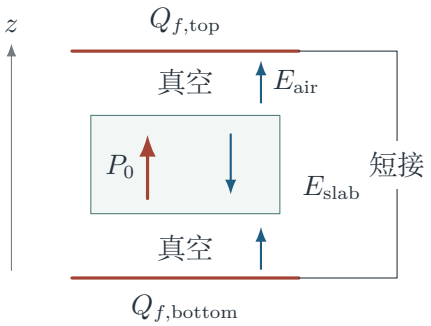


图 14-40　题图　永久极化片两侧还留有真空间隙



短接：$D = P_0 t/d$

$E_{\mathrm{slab}} < 0,\ E_{\mathrm{air}} > 0$

图 14-58　解答分析图　短接只要求各区域电压之和为零

14-T9　形状改变后的极化强弱

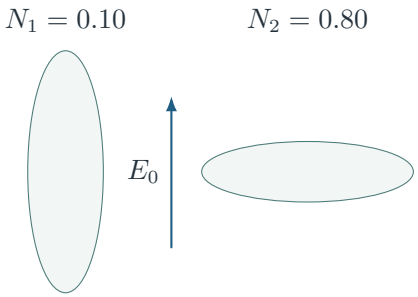


图 14-41　题图　两物体体积相同，图中形状用于表示退极化强弱

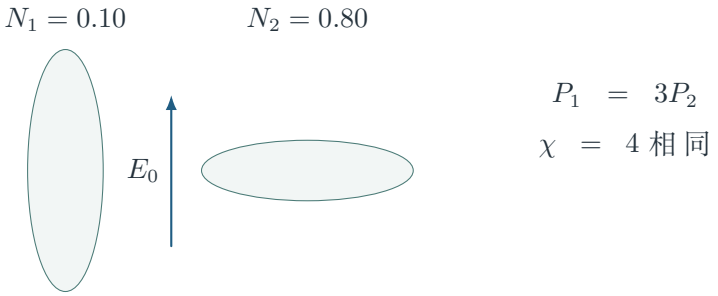


图 14-59　解答分析图　材料参数相同，极化差来自几何反馈

14-T10　各向异性平板中的场偏转

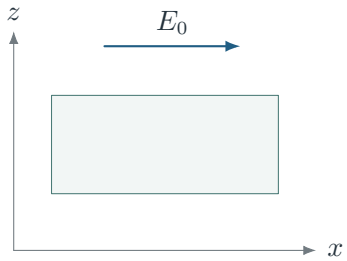
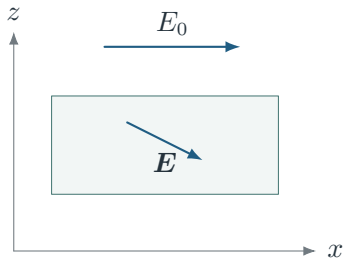


图 14-42　题图　介电张量有非对角项，外场仍平行板面



$$E_z = -E_0/2$$
$$D_z = 0$$
$$P_z = \varepsilon_0 E_0/2$$

图 14-60　解答分析图　面电荷产生法向场，电位移法向保持连续

14-T11　局域场修正的数值大小

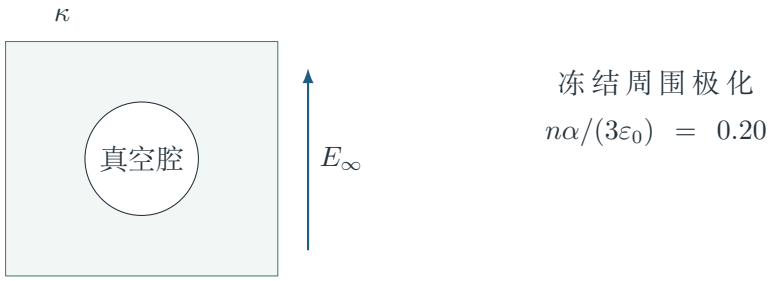


图 14-43　题图　球腔表示洛伦兹局域场构造，不作真实腔再极化

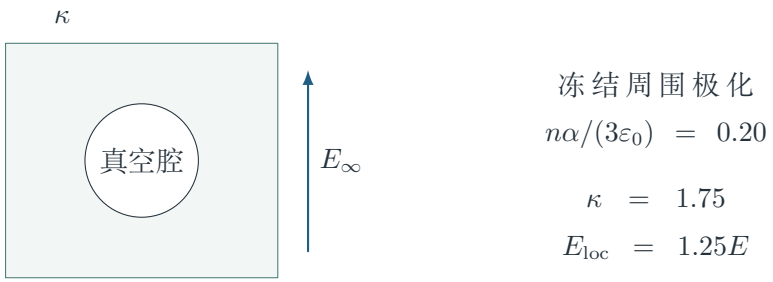


图 14-61　解答分析图　极化反馈提高分子所受场和宏观电极化率

14-T12　由温度变化分离两种极化

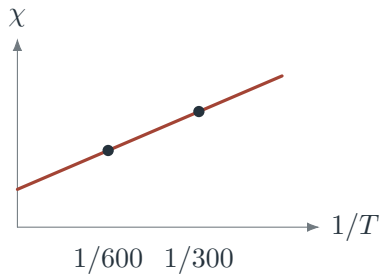


图 14-44　题图　固定分子数密度后，电极化率对倒温度线性

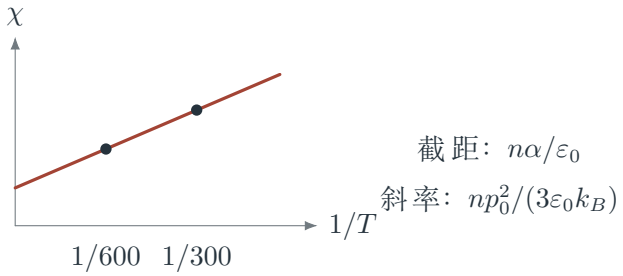


图 14-62　解答分析图　两个温度点分别确定截距和斜率

14-T13　电畴壁上的面电荷

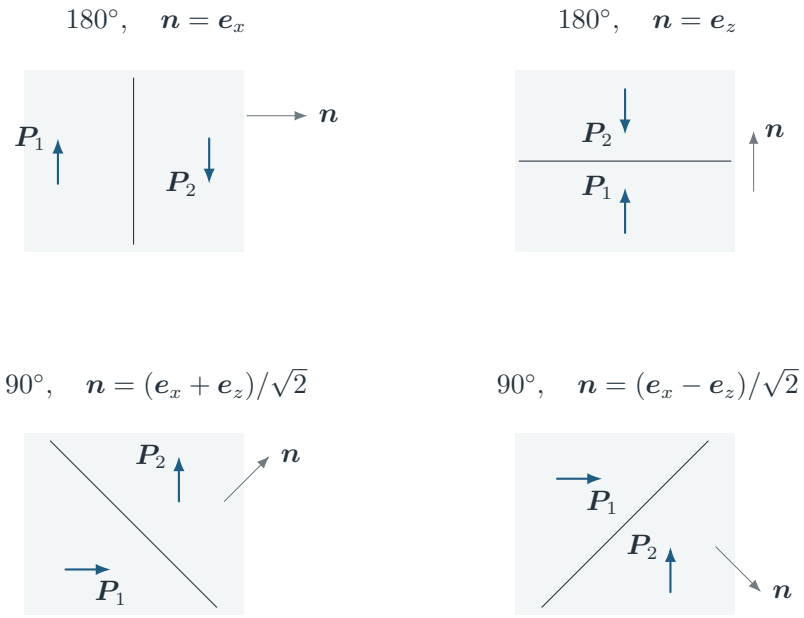


图 14-45　题图　在同一 x-z 坐标约定下并列题设四种极化与壁法向

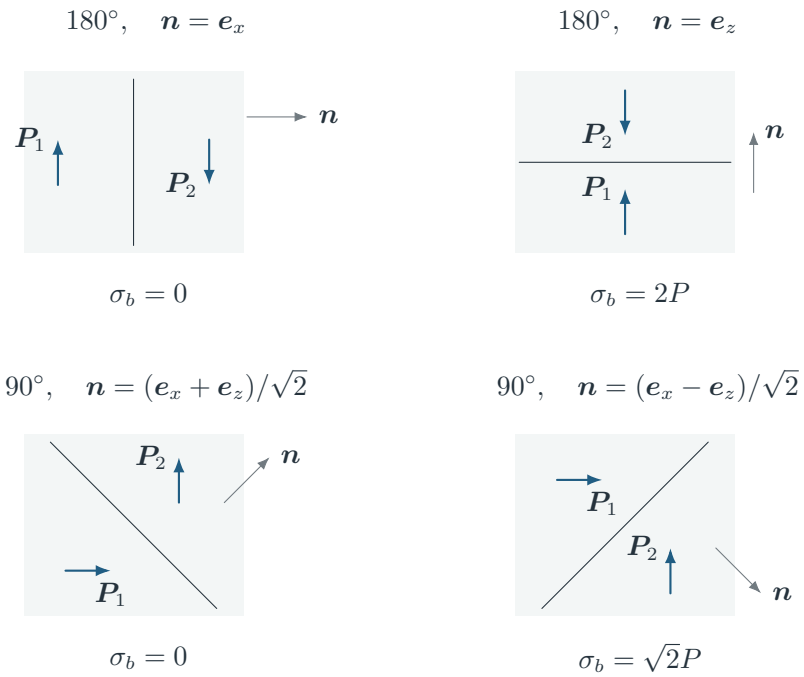


图 14-63　解答分析图　逐图投影极化差，法向改变可使同类畴壁带电或不带电

14-T14　极化滞后时的周期损耗

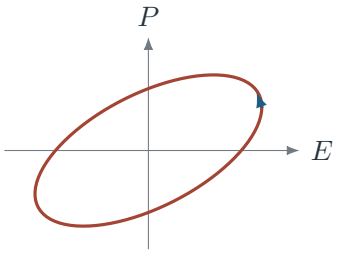
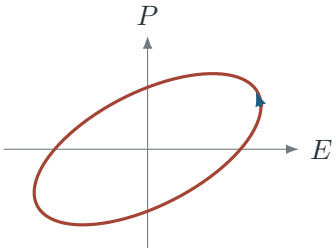


图 14-46　题图　动态相位滞后使升场与降场轨迹分开



$$w_{\text{loss}} = \oint E\,\mathrm{d}P$$
$$= \pi E_m P_m \sin\delta$$

图 14-64　解答分析图　回线按实际循环方向围成正的周期损耗

14-T15　极化球中的偏心空腔

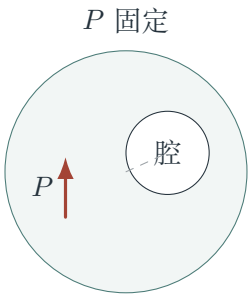
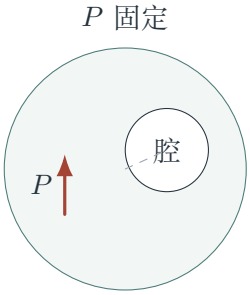


图 14-47　题图　空腔偏离大球中心，但完全包含在大球内



大球内场：$-P/(3\varepsilon_0)$

移去小球：$+P/(3\varepsilon_0)$

图 14-65　解答分析图　两个均匀极化球的内场在腔内抵消

14-T16　另一种介质中的小球

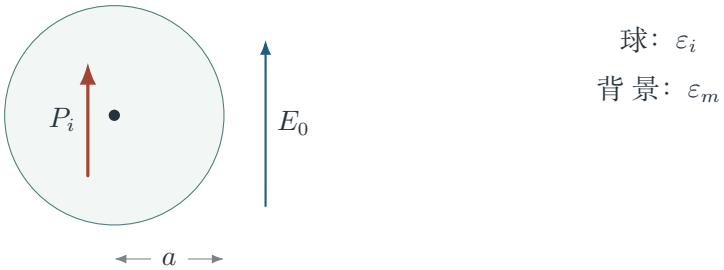


图 14-48　题图　内部材料极化顺场，外部扰动由介质差决定

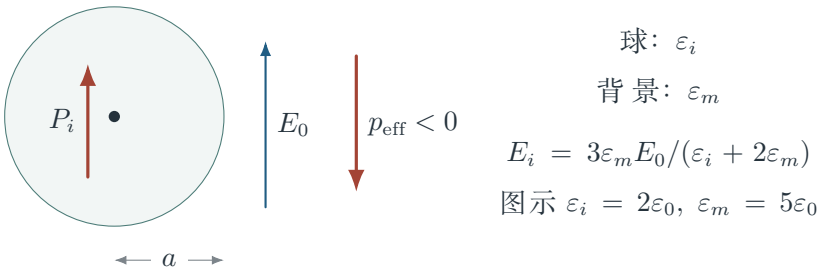


图 14-66　解答分析图　给定数值下扰动偶极矩反向，球内部的材料极化仍顺场

14-T17　综合题：包覆介质球的极化

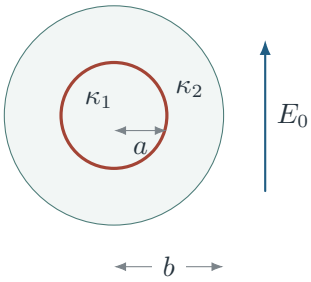
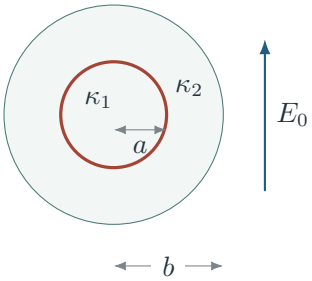


图 14-49　题图　核心与外壳分别具有不同介电常数



$$
\begin{aligned}
r &= a\text{：}\ \varphi, D_n \text{ 连续} \\
r &= b\text{：}\ \varphi, D_n \text{ 连续} \\
p &= 4\pi\varepsilon_0 G
\end{aligned}
$$

图 14-67　解答分析图　四个边界条件同时确定核心场和外部偶极项

14-T18　综合题：非均匀场中的椭球运动

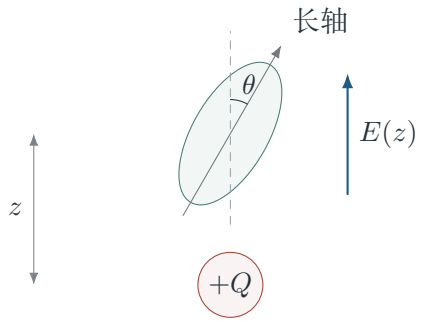
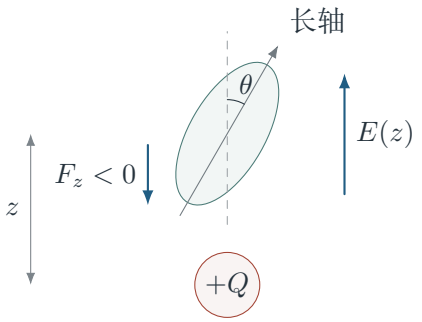


图 14-50　题图　小椭球处于点电荷的局部非均匀场，尺寸远小于距离



$$U_{\text{eff}} = -\mathcal{A}(\theta)E(z)^2/2$$

$$\theta = 0, \pi \text{ 稳定}$$

图 14-68　解答分析图　极化率较大的轴趋于顺场，净力趋向高场区域